Week 26: Same area, different boundaries

Adult guide built from the problem files · Unpiloted. Physical pretests unperformed.

Each problem's key repeats the main answer from the print guide, week-26-facilitator.pdf, whose source is editable/facilitator-src/build_guide.py; the arguments, held hints and figures are on the guide page named in the key's first line. The overview repeats the guide's unnumbered overview. Change both together; check_guide_copy.py checks that every sentence here is in the guide.

Mathematical overview

The mathematical core

Use $n \geq 1$ unit squares, without overlap, connected by full shared sides. Count all exposed unit edges, including hole edges. If e is the number of shared sides, then $\mathbf{P = 4n - 2e}$: each meeting removes two edge contributions. Hence every perimeter is even. Adding a tile touching k old sides changes P by $4 - 2k$.

Maximum: Connectivity gives $e \geq n - 1$, so $P \leq 2n + 2$; a straight row attains it. Equality holds exactly when the tile-adjacency graph is a tree. This rules out a 2-by-2 block. Whether it rules out a hole depends on corners. The Grades 4–5 page says two tiles touching at a corner leave no gap; then a tree can enclose a hole without adding a shared side. If instead a corner gap let the inside out, a hole would need a ring of side-joined tiles, a cycle, so no tree could have one (page 14).

Minimum: If r rows and c columns are occupied, then $P \geq 2(r+c)$ and $n \leq rc$. For $P = 2s$, the largest possible area is floor$(s/2)$·ceil$(s/2)$, attained by a balanced rectangle. For fixed n, the minimum perimeter is 2 ceil$(2\sqrt{n})$. Near-square rectangles with corner-started partial strips attain the bound for every n. Equality in $P \geq 2(r+c)$ requires one occupied interval in every occupied row and column, not necessarily a full rectangle.

What the activity develops

K–1: Build legal shapes, count exposed sides and compare records. The shared launch models both a whole-side join and boundary counting with three tiles.

Grades 2–3: Track shared-side cancellation and one-tile changes, then explain a longest-boundary construction.

Grades 4–5: Pair row/column lower bounds with attaining shapes and generalize exact area–perimeter limits. Square-root notation is an optional adult shorthand.

From trials to explanation

A best shape found so far is a record and a candidate optimum. A conjecture predicts it cannot be beaten. A matching lower or upper bound, together with a construction attaining it, proves the optimum. The detailed counting arguments remain on printed page 3.

Problems and routes

Problem	K–1	2–3	4–5	route
Every four-tile shape and its boundary	P1			first
Shortest and longest with 5 and 6 tiles	P2			first
Shortest and longest with 7 and 8 tiles	P3			first
Six tiles with 10, 12 and 14 boundary sides	P4			return
Which boundaries five tiles can make	P5			return
Move one tile to shorten the boundary	P6			return
Shortest and longest eight-tile shapes, two of each		P1		first
Every boundary change from adding one tile		P2		first
Shared sides and a rule for the boundary		P3		first/return
Longest boundary with 4, 7, 10 and 12 tiles		P4		first
Which eight-tile shapes have the longest boundary?		P5		return
Fewest starting tiles for each boundary change		P6		return
Shortest boundary with 7, 10 and 13 tiles, two ways			P1	first
Longest twelve-tile boundary: blocks and holes			P2	return
Rows, columns and boundary			P3	first/return
Most tiles inside a given boundary			P4	first/return
Shortest boundary for 12, 13, 17, 20 and 21 tiles			P5	first/return
A rule for the greatest area, and 37, 50 and 73 tiles			P6	return

First route. K–1 P1–P3; Grades 2–3 P1–P4; Grades 4–5 P1, P3–P5. **Return visits.** K–1 P4–P6; Grades 2–3 P3, P5–P6; Grades 4–5 P2–P6.

Key, one entry per problem

Every four-tile shape and its boundary

K–1 P1 · first route

Idea. Up to turns and flips there are five four-tile shapes; the square has boundary 8 and the other four 10, because each shared side hides two edges: $P = 4n - 2e$.

Print guide, p. 4. **Exact answer:** five shapes up to turns and flips. The square has 8 exposed sides; the other four have 10. The sixth blank recording grid is not a request for a sixth shape.

For the counts, the square has four shared sides, giving $16 - 8 = 8$. Each other shape has three shared sides, giving $16 - 6 = 10$. Let children check with markers before introducing the subtraction shorthand.

Shortest and longest with 5 and 6 tiles

K–1 P2 · first route

Idea. With 5 and 6 tiles the shortest boundary is 10, since boundary 8 fits only a 2-by-2 box, and the longest is $2n + 2$: 12 and 14.

Print guide, p. 5. **Exact extrema:** 5 tiles give 10 and 12; 6 give 10 and 14; 7 give 12 and 16; 8 give 12 and 18. These are witnesses, not lists of every optimizer.

Longest: the connection argument on page 3 gives $2n + 2$. The rows shown meet that bound. Children can start with 4 sides on one tile and add 2 exposed sides for each new tile in a row.

Shortest for 5 or 6: boundary 8 or less would allow $r + c$ at most 4. The largest rectangle with that sum is 2 by 2, holding only 4 tiles. Thus at least 10 sides are needed, and the diagrams attain 10.

Shortest and longest with 7 and 8 tiles

K–1 P3 · first route

Idea. With 7 and 8 tiles the shortest boundary is 12, since boundary 10 fits only a 2-by-3 box, and the longest is $2n+2$: 16 and 18.

Print guide, p. 5. **Exact extrema:** 5 tiles give 10 and 12; 6 give 10 and 14; 7 give 12 and 16; 8 give 12 and 18. These are witnesses, not lists of every optimizer. In particular, the seven-tile minimum is a 2-by-4 rectangle with one corner missing.

Longest: the connection argument on page 3 gives $2n + 2$. The rows shown meet that bound.

Shortest for 7 or 8: boundary 10 or less would allow $r + c$ at most 5. The best such box is 2 by 3, holding only 6 tiles. At least 12 sides are needed, and the diagrams attain 12. Odd boundary lengths are excluded by paired shared-edge cancellations.

Six tiles with 10, 12 and 14 boundary sides

K–1 P4 · return route

Idea. Boundary 10 with six tiles forces the full 2-by-3 rectangle, so only 12 and 14 can be made in two different ways.

Print guide, p. 6. Exactly 12 and 14 have two different answers.

With six tiles, $P = 10$ occurs only for a 2-by-3 rectangle up to turns/flips. Indeed $P = 10$ implies $r+c$ at most 5. To hold 6 cells, dimensions must be 2 and 3 and every cell of that box must be filled. Therefore the second 10-side answer space cannot contain a different shape. The 12-side examples above have different bounding boxes (4 by 2 and 3 by 3); the two 14-side examples are a row and a bent row.

Which boundaries five tiles can make

K–1 P5 · return route

Idea. Every boundary is even, because each shared side hides two edges, and boundary 8 holds at most four tiles, so five tiles make only 10 and 12.

Print guide, p. 6. Among 8, 9, 10, 11, 12, 13, only 10 and 12 work.

Boundary 8 would force a box holding at most 4 tiles (page 5), so it is too short. Odd lengths 9, 11 and 13 are impossible because starting from 4 sides per tile, every shared side removes two. Thus every boundary is even. The two drawings attain the remaining candidates. Equivalently, every occupied row contributes an even number of vertical boundary edges and every occupied column an even number of horizontal edges.

Move one tile to shorten the boundary

K–1 P6 · return route

Idea. One move takes the middle start to the 2-by-3 rectangle (10) and the L to 12, which is best because 10 needs a 2-by-3 rectangle; the straight row cannot get shorter than 14.

Print guide, p. 7. **Exact answer:** only the middle and bottom starts can be made shorter. Their best boundaries are 10 and 12; the straight row stays at 14.

Moves shown: in the top row, move the right endpoint just below the left endpoint. In the middle start, move the rightmost tile of its four-tile row into the gap between the two tiles in the next row. In the bottom L, move the far end of its four-tile arm into the inside corner.

Shortest and longest eight-tile shapes, two of each

Grades 2–3 P1 · first route

Idea. Eight tiles have shortest boundary 12 (the 2-by-4 rectangle or the 3-by-3 square missing a corner) and longest $2 \cdot 8 + 2 = 18$ (any shape with exactly 7 shared sides).

Print guide, p. 8. **Exact answer:** minimum 12, maximum 18. The 2-by-4 rectangle and 3-by-3 square missing a corner are different shapes, because their bounding boxes are different even after turning. The two long-boundary examples are also different.

For a short boundary, P at most 10 forces $r + c$ at most 5, whose largest possible box is 2 by 3, holding 6 tiles. Eight tiles therefore need P at least 12, and both compact figures attain it. For the maximum, connectivity requires at least 7 shared sides; $32 - 2 \times 7 = 18$. Both long examples have exactly 7 shared sides.

Every boundary change from adding one tile

Grades 2–3 P2 · first route

Idea. A new tile beside k old tiles changes the boundary by $4 - 2k$, so the four starts allow {+2}, {0, +2}, {-2, +2} and {-4, +2}.

Print guide, p. 9. For +2 in every start, attach a tile beyond the left end of the top row. The resulting boundaries are 14, 14, 14 and 18 in reading order. Each start allows at most two changes and the row only one, so at least one recording box per start stays blank.

An addition touching k old neighbors changes P by $4 - 2k$. The row has no empty position touching two of its cells. The stair has empty positions touching one or two cells, but none touching three. In the U, its central gap touches three; all other available positions touch only one (there is no two-neighbor position). In the frame, the center touches four and every exterior position touches one. This accounts for every empty side-neighbor position, so the complete change sets are {+2}, {0,+2}, {-2,+2}, {-4,+2}.

The frame has outer boundary 12 and inner boundary 4, totaling 16. Filling the hole removes its 4 inner edges, giving 12. Counting only the outside would give a wrong starting value and obscure the intended phenomenon.

Shared sides and a rule for the boundary

Grades 2–3 P3 · first/return route

Idea. Each shared side hides two tile edges, so the boundary is $4n - 2e$: ten tiles with 9 to 13 shared sides have boundaries 22 down to 14.

Print guide, p. 10. Each figure has ten tiles. The two-row examples have row lengths (8,2), (7,3), (6,4) and (5,5), all left-aligned. Their shared sides are, respectively, $7 + 1 + 2 = 10$; $6 + 2 + 3 = 11$; $5 + 3 + 4 = 12$; and $4 + 4 + 5 = 13$. These sums count within each row, then between rows. The row of ten has 9 meetings.

Rule: four times the tile count, minus twice the shared-side count. In symbols, $P = 4n - 2e$. The proof is the two hidden contributions at each meeting (page 3). This identity includes inner hole edges and allows shapes with holes.

Longest boundary with 4, 7, 10 and 12 tiles

Grades 2–3 P4 · first route

Idea. A connected shape of n tiles has at least $n - 1$ shared sides, so its boundary is at most $2n + 2$: 10, 16, 22 and 26.

Print guide, p. 10. Maxima 10, 16, 22, 26.

A connected shape needs at least $n - 1$ meetings, so $P = 4n - 2e$ is at most $2n + 2$. The rows attain it. Explain the connection bound by discovering tiles one at a time through the already connected part, assigning each new tile a

distinct joining edge. Extra meetings can only make the boundary shorter. This proves the maximum for every n, not just these four trials.

Which eight-tile shapes have the longest boundary?

Grades 2–3 P5 · return route

Idea. An eight-tile shape has the longest boundary, 18, exactly when it has 7 shared sides, so branches and bent paths qualify and the rectangle and the ring do not.

Print guide, p. 11. In worksheet reading order the boundaries are **18, 18, 12, 16**. The straight and branching examples each have 7 shared sides. The rectangle has 10; the frame has 8. Since 8 tiles allow at most 18 exposed sides, exactly the first two reach the maximum.

Every eight-tile connected shape has at least seven meetings. A chosen example with exactly seven is automatically maximal, because $32 - 14 = 18$. Therefore a child does not have to compare it with every conceivable eight-tile shape. A branch is allowed: the key is the number of meetings, not whether the shape looks like a single path.

Fewest starting tiles for each boundary change

Grades 2–3 P6 · return route

Idea. A change of $4 - 2k$ needs k tiles around an empty cell, all connected without it, so $+2$, 0, -2 and -4 need at least 1, 3, 5 and 7 starting tiles.

Print guide, p. 12. **Exact minima:** 1, 3, 5 and 7 starting tiles. The starting perimeters are 4, 8, 12 and 16; after addition they are 6, 8, 10 and 12. The last start is a 3-by-3 frame missing its center and one corner. It is side-connected and valid; eight starting tiles are not necessary.

+2 needs one neighbor. A starting shape is nonempty, so one is the lower bound and the singleton works.

0 needs two neighbors. Any two side-neighbor cells of the same empty target are not side-adjacent to one another. Two starting tiles therefore cannot form a connected shape. Three arranged as an L do work.

-2 **needs three neighbors.** These three neighbor cells are pairwise nonadjacent. One additional cell other than the target can touch at most two of them: only a diagonal corner can touch two, and no allowed cell touches three. Thus four starting tiles cannot connect all three target neighbors. Five tiles in the U shown do work.

-4 **needs four neighbors.** Color the grid like a chessboard, with the empty target black. All four required neighbors are white. With at most six starting tiles there are at most two other tiles. A black connector other than the target touches at most two of the four required neighbors. If both extra tiles are black, they are not adjacent to one another and give at most four total shared sides. Six connected tiles need at least five. If at most one extra tile is black, at most two required neighbors can meet it; no other white required neighbor has a connection. Five or fewer starting tiles fail for the same reason. Seven in the pictured almost-frame work.

Shortest boundary with 7, 10 and 13 tiles, two ways

Grades 4–5 P1 · first route

Idea. A boundary of $2s$ holds at most $\lfloor s/2 \rfloor \lceil s/2 \rceil$ tiles, so 7, 10 and 13 tiles need at least 12, 14 and 16, and compact shapes reach them.

Print guide, p. 13. **Exact minima:** 7 gives 12, 10 gives 14, 13 gives 16. Each figure has every row and every column in one interval, so $P = 2(r + c)$.

At boundary 10, the biggest possible area is $2 \times 3 = 6$, so 7 needs at least 12. At boundary 12, the biggest area is $3 \times 3 = 9$, so 10 needs at least 14. At boundary 14, the biggest area is $3 \times 4 = 12$, so 13 needs at least 16. The diagrams attain these next even lengths.

Longest twelve-tile boundary: blocks and holes

Grades 4–5 P2 · return route

Idea. Twelve tiles have boundary at most 26, reached exactly when there are 11 shared sides (a tree); a 2-by-2 block makes a cycle, but with tiles touching at a corner closing the gap, as the page says, a tree can enclose a hole.

Print guide, p. 14. **Exact answer:** maximum 26. All three examples attain it, with twelve tiles and eleven shared sides. None is a straight row. The connection bound gives P at most $48 - 22 = 26$, so no longer boundary is possible.

A 2-by-2 block: no. A hole: yes, sealed at a corner.

A 2-by-2 block creates a four-edge cycle in the tile-adjacency graph. Removing one edge from a cycle leaves its vertices connected, so any connected graph with such a cycle has at least n edges rather than $n - 1$. Its perimeter is at most $2n$, below the maximum. This rules out the block.

In the third drawing, the empty center cell has a tile on each of its four sides, and the empty upper-left corner cell is outside. Where those two empty cells meet diagonally, two tiles meet at a corner and close the passage, so in the union of closed squares the center is a hole. Four of the 26 exposed edges surround it; the outer boundary contributes 22.

This corner contact adds no shared side, so the shape is still a tree: “tree adjacency implies no holes” is false once two tiles meeting at a corner close a gap, and the student page says they do. A child who answers no is usually thinking of a hole closed by full sides. That argument is right for such a hole: it needs a ring of side-joined tiles, a cycle, which forces at least n shared sides. Point to the page’s sentence about corners and ask whether the ring can be left open at one corner.

Rows, columns and boundary

Grades 4–5 P3 · first/return route

Idea. A shape spanning r rows and c columns has boundary at least $2(r + c)$, with equality exactly when every row and column is one run, so 4 rows and 5 columns force at least 18.

Print guide, p. 15. All three supplied shapes have twelve tiles. Their exact (rows, columns, boundary) triples, left to right, are **(2,6,16), (3,4,14), (4,5,20)**. The third is the second construction below, so a child may copy it as one of the two shapes the next question asks for.

For the second construction, move the right tile from the bottom row of the first construction to the far right of the second row. Its rows become 5; 3 plus a separated tile; 2; 1. It remains connected through the full top row. Both examples use twelve tiles and exactly four rows and five columns.

Each of four occupied rows contributes at least two vertical boundary edges, for at least 8. Each of five occupied columns contributes at least two horizontal boundary edges, for at least 10. These count disjoint edge directions, so P is at least 18. The first construction attains it. Its rows and columns are all contiguous, so it has exactly those mandatory boundary edges.

The second construction has an extra gap in its second row, producing two extra vertical boundary edges; its columns remain contiguous. Thus $P = 20$.

Most tiles inside a given boundary

Grades 4–5 P4 · first/return route

Idea. Boundary $2s$ allows $r + c \le s$ and so at most $\lfloor s/2 \rfloor \lceil s/2 \rceil$ tiles: 9, 12, 16 and 20 for boundaries 12, 14, 16 and 18.

Print guide, p. 16. The exact answers are **9, 12, 16, 20**. For boundary 12, 14, 16 or 18, the sum $r + c$ cannot exceed 6, 7, 8 or 9. For each sum, the balanced dimension pair gives the largest product (page 3). A smaller sum cannot give more area: increasing one positive dimension increases its product. The displayed rectangles reach the bounds with exactly the requested boundary.

Shortest boundary for 12, 13, 17, 20 and 21 tiles

Grades 4–5 P5 · first/return route

Idea. The least boundaries for 12, 13, 17, 20 and 21 tiles are 14, 16, 18, 18 and 20, each certified by the greatest area for the previous even boundary.

Print guide, p. 16. The exact answers are **14, 16, 18, 18, 20**. Construction row lengths are (4,4,4), (4,4,4,1), (4,4,4,4,1), (5,5,5,5), and (5,5,5,5,1). Every row/column is contiguous, so its perimeter is twice its row-plus-column span.

Lower-bound certificates: boundary 12 holds at most 9 tiles, excluding it for $n = 12$. Boundary 14 holds at most 12, excluding it for $n = 13$. Boundary 16 holds at most 16, excluding it for $n = 17$ and $n = 20$. Boundary 18 holds at most 20, excluding it for $n = 21$. Because boundaries are even, each construction at the next even value is optimal. Bounds for smaller perimeters are no larger, so excluding the preceding even value excludes all smaller ones.

A rule for the greatest area, and 37, 50 and 73 tiles

Grades 4–5 P6 · return route

Idea. The greatest area inside boundary $2s$ is $\lfloor s/2 \rfloor \lceil s/2 \rceil$, so the least boundary for n tiles is $2\lceil 2\sqrt{n}\,\rceil$: 26, 30 and 36 for 37, 50 and 73.

Print guide, p. 17. **Rule:** halve the even boundary P to get s. Split s into two positive whole numbers as equal as possible, $a = \lfloor s/2 \rfloor$, $b = \lceil s/2 \rceil$. The greatest area is ab.

Every shape has area n at most rc and $r + c$ at most s. If $r + c$ is smaller, increase a dimension until the sum is s; the product increases. With sum s fixed, replacing (r, c) by $(r + 1, c - 1)$ when c is at least $r + 2$ increases the product by $c - r - 1$. Repeat until the dimensions differ by at most one. This proves n at most ab. The full a-by-b rectangle has area ab and boundary $2(a + b) = P$, so the bound is attained. P at least 4 makes both dimensions positive.

37: boundary 24 permits at most $6 \times 6 = 36$. The pictured 6-by-6 square plus one tile has 37 and boundary $24+2 = 26$.
50: boundary 28 permits at most $7 \times 7 = 49$. The pictured 7-by-7 square plus one tile has 50 and boundary $28+2 = 30$.
73: boundary 34 permits at most $8 \times 9 = 72$. The pictured 8-by-9 rectangle plus one tile has 73 and boundary $34 + 2 = 36$.

All three added tiles meet the rectangle on exactly one side. Each next even boundary is attained; the preceding one and all smaller values are impossible. Thus the exact minimum boundaries are **26, 30, 36**.